

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 02

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• UV light splits diatomic oxygen (O₂) into two monatomic oxygen atoms (O) • Monatomic oxygen reacts with diatomic oxygen to form ozone (O₃) • UV light also splits ozone into O₂ and O; the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat)	3
01.2	• Useful: boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants); dissolve grease but do not damage electrical components (solvents); non-flammable; non-toxic • Persistence: chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere • Dissociation by UV: in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals • Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation; chlorine is regenerated so one atom destroys many ozone molecules; bromine / iodine act similarly	6
01.3	• UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) • Damage to leaf tissue and reduced photosynthesis / crop yields (plants) • Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs	3
01.4	• An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution	1
01.5	• Concentration varies with altitude and the stratosphere is not normally sampled; Dobson Units measure the total column thickness instead	1
01.6	• The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
02.1	• Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients; worms create tunnels improving drainage and aeration • Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds • Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions; nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption • Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake	6
02.2	• This is the range of tolerance of most plants and soil biota; acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes • In alkaline soils phosphates become insoluble / unavailable	2
02.3	• Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate • Infiltration falls so waterlogging is more likely and runoff / erosion increase	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	
03.1	• A waste is assumed to cause pollution until research shows it is unlikely to do so	1
03.2	• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply	2
03.3	<i>Any 2 from:</i> • Marine discharges where currents dilute and disperse emissions • Emissions downwind of urban areas • Not discharging waste onto permeable rock above an aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
04.1	• The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean • The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly	2
04.2	• t-test • Comparing two means of measured (continuous) data	2
04.3	• There is a 30% probability that the difference occurred by random chance • Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted	2
04.4	• Two correlated variables may both be caused by a third variable that was not investigated • Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation	2
04.5	• Increase the number of replicates (reduces the effect of variability) • Standardise other variables (soil, water, light) more closely to reduce scatter	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
05.1	- Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium - Positive feedback: a change causes other changes that increase the original change	2
05.2	- Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced - Increased photosynthesis: higher temperatures and CO₂ increase photosynthesis, removing CO₂; if stored in woody tissue atmospheric CO₂ rises less	4
05.3	- Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground - Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming - Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature - If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming	4
05.4	<i>Any 3 from:</i> - Reduced albedo as snow / ice melts - Ocean acidification reducing coral survival and carbon sequestration - Increased forest and peat fires releasing CO₂ - Faster decomposition of soil organic matter releasing CO₂ - Increased water vapour (a greenhouse gas) in warmer air - Increased cirrus (high) cloud which traps infrared	3
05.5	- A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable - So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
06.1	<i>Any 3 from:</i> - Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) - Very complex, expensive technology needing an advanced industrial infrastructure - Concerns about links between civil nuclear materials and weapons / terrorism - Uncertainty over permanent disposal of radioactive waste - Uncertain decommissioning costs	3
06.2	- Only 0.7% of natural uranium is fissile U-235; the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) - Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) - So much more energy is harnessed from the mined uranium; fast reactors need no moderator but are complex and costly	3

Question	Marking guidance	Marks
06.3	• Advantages: thorium is three times more abundant; much less radioactive waste with shorter half-lives; harder to make weapons material; no fuel enrichment needed • Disadvantages: thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate); U-233 is an alpha emitter so very hazardous to handle; less developed technology with high development costs	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	
07.1	• B - Tropospheric ozone (formed by reactions of NO _x in sunlight)	1
07.2	• Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO_x from high-temperature combustion • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO_x dissolve to form nitric acid • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO_x photochemistry	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
08.1	• B - rivers are accessible over a large area with few natural contaminants	1
08.2	• Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals	2
08.3	• Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
09.1	• Aerobic decomposition produces little methane and returns nutrients to the soil	1
09.2	• It may contain concentrated heavy metals / toxic residues • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble	2
09.3	<i>Any 2 from:</i> • Impermeable polymer / clay liner to stop leachate escaping • Collection and treatment of leachate • Collection of methane and use as fuel • Daily covering of waste / final capping and restoration	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
10.1	• Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) • They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone • The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface	3
10.2	<i>Any 2 from:</i> • Provides carbon, oxygen, hydrogen and nitrogen (as N₂, O₂, CO₂, H₂O) for biological molecules • Delays the escape of infrared energy, keeping the surface warm • Distributes heat and water vapour by winds / ocean currents • Provides atmospheric pressure so liquid water can exist • Deflects damaging radiation of the solar wind in the upper atmosphere	2
10.3	• Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers	1

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Question	Marking guidance	Marks
11.1	Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Nature of the problem: interconnected atmosphere-hydrosphere-biosphere systems; changes over different time and spatial scales; natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change; time delay between cause and effect (ocean heat capacity) • Limitations of historic data: sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • Proxy data and their reliability: dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • Modern monitoring: long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty; ecological uncertainty (species survival, distribution, fragmentation) • Computer models: how they are tested and refined; importance of accurate representative data; scenario-based predictions and ranges • Uncertainty over ENSO / Conveyor / jet stream responses • Evaluation and conclusion: confidence in the overall trend and human cause despite detailed uncertainty; role of peer review and international bodies (IPCC) Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	25

Question	Marking guidance	Marks
11.2	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25